NP - MIT

P = Problems that can be solved in polynomial time ($n^{O(1)}$)
(size n)

NP = ~~Problem~~ {decision problems solvable in nondeterministic polynomial time}

answer is YES or NO

Can guess one out of polynomially many options in $O(1)$ time.

→ if any guess would lead to a YES, then we get such a guess.

Example:

3SAT: Given Boolean formula of the form

$$(x_1 \vee x_3 \vee \bar{x}_6) \wedge (\bar{x}_2 \vee x_3 \vee \bar{x}_7)$$

literals

Clause

Can you set the variable $x_1, x_2 \ldots$ → {T, F} such that the formula comes out as T (True)

Satisfying assignment

NP: Guess whether $x_1 = T$ or F

guess $x_2 = T$ or F

⋮

Check formula → T → return YES

F → return NO

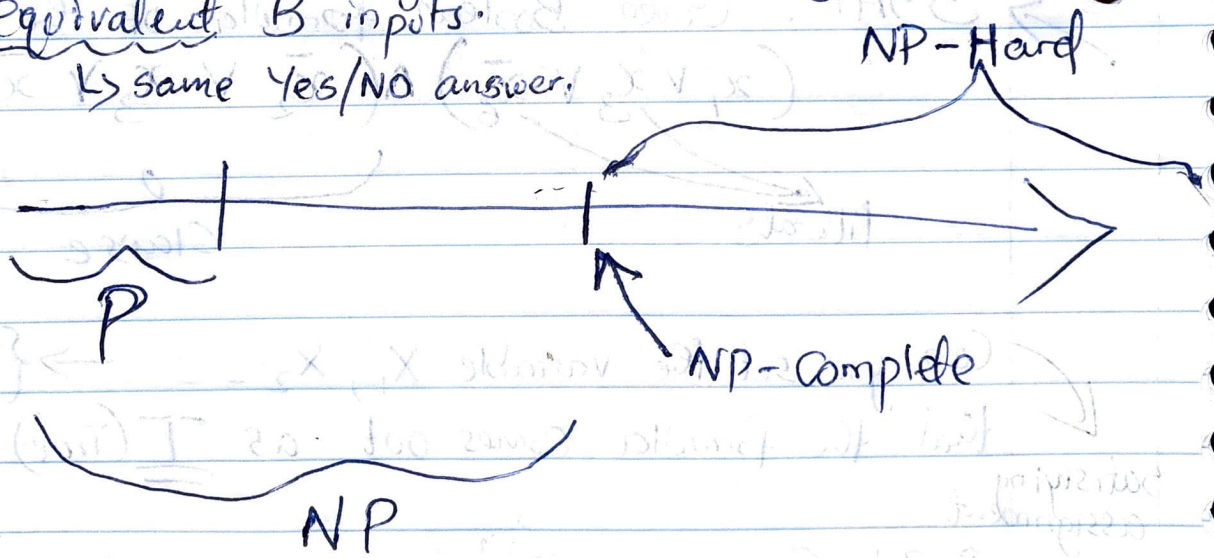
NP = { decision problems with poly-size certificates & poly-time verifiers for YES inputs }

→ X is <u>NP-complete</u> if $X \in NP$ & NP-hard.

→ X is <u>NP-hard</u> if every problem $Y \in NP$ reduces to X

→ <u>Reduction</u> from problem A → problem B = poly-time algorithm converting A inputs → equivalent B inputs.

↳ same Yes/NO answer.



⟹ $x \notin P$ unless $P = NP$

→ If $B \in P$ then $A \in P$

→ If $B \in NP$ then $A \in NP$

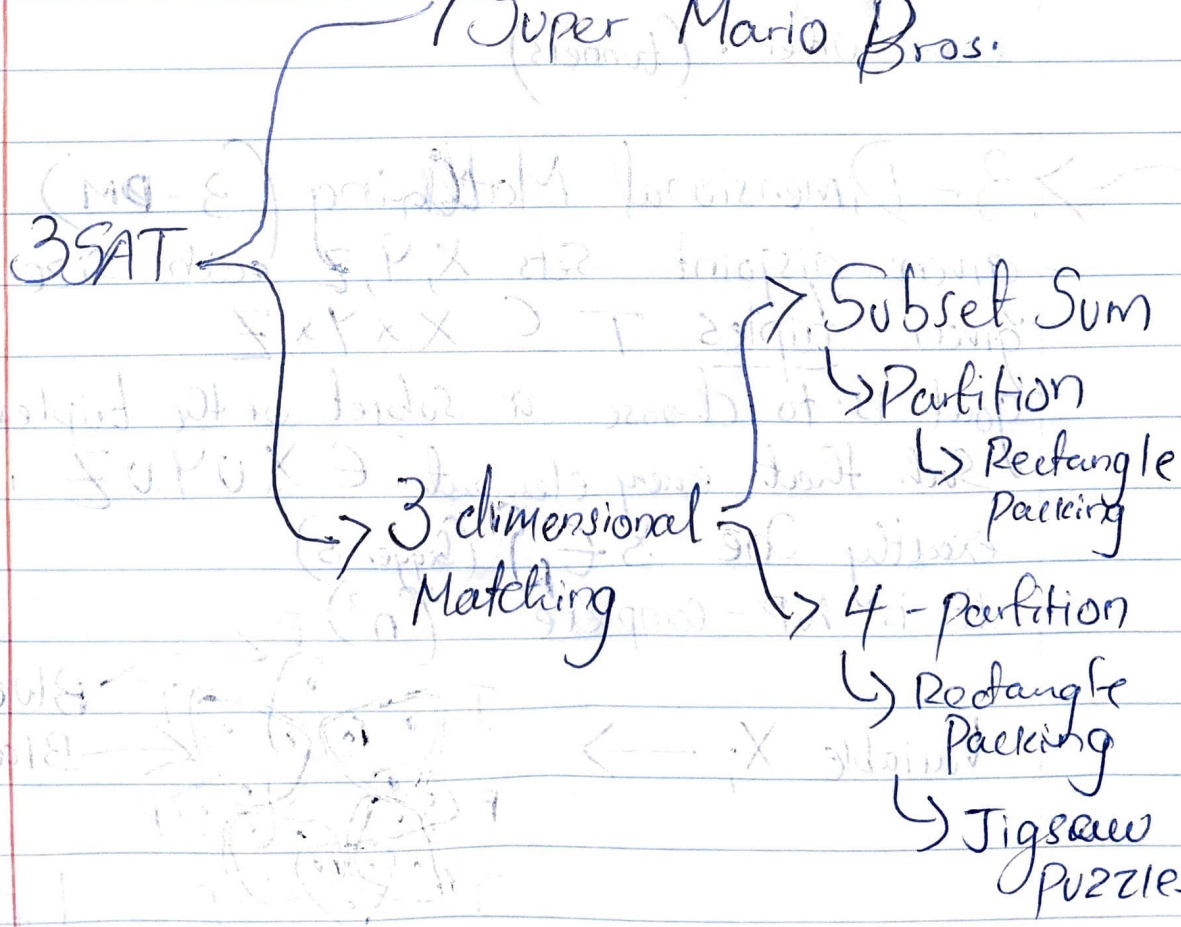
Guide on how to Prove:

How to prove X is NP-Complete:

1. Prove $x \in NP$ → non-det. algorithm / → certificate + verifier
2. Prove $x \in NP$ hard

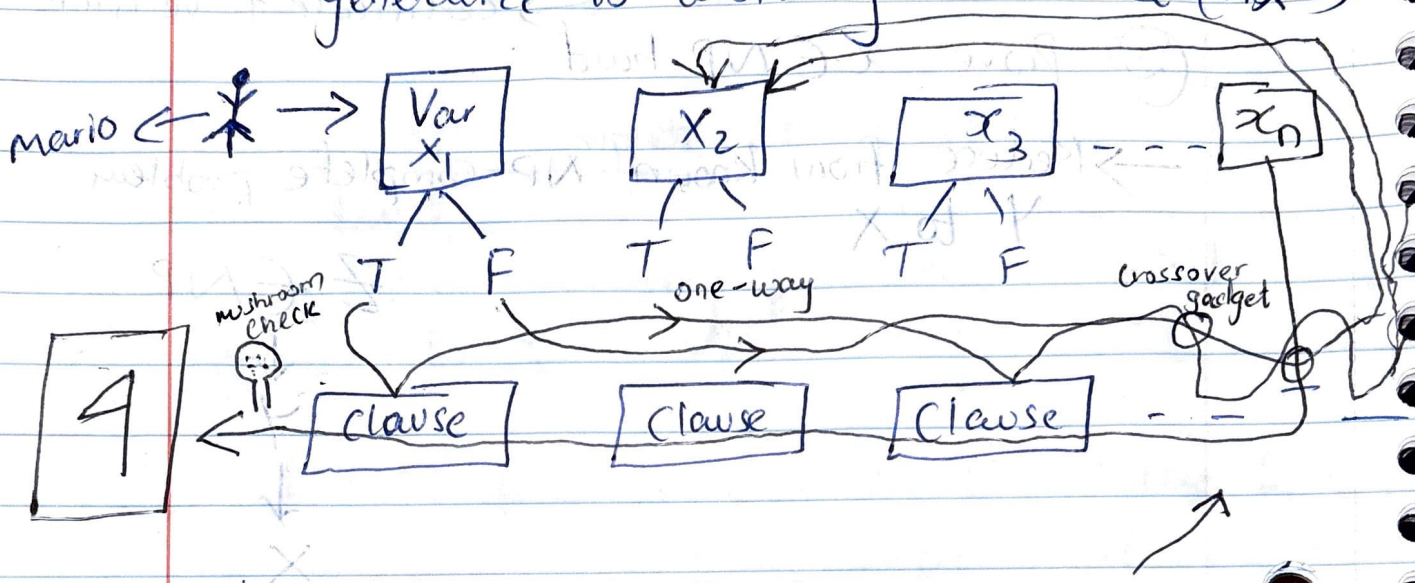
→ Reduce from known NP-Complete problem Y to X

$Z \in NP$
↓
Y
↓
X



→ Super Mario Bros : is NP-hard

- generalize to arbitrary Screen size ($n \times m$)



* The problem is that the wires cross each other. (tunnels)

→ 3-Dimensional Matching (3-DM)

- given disjoint sets X, Y, Z each size n
- given triples $T \subseteq X \times Y \times Z$
- goal is to choose a subset of the triples $S \subseteq T$ such that every element $\in X \cup Y \cup Z$ is in exactly one $s \in S$ (bigger S)
- It is NP-complete (n^3)

Variable X_i →

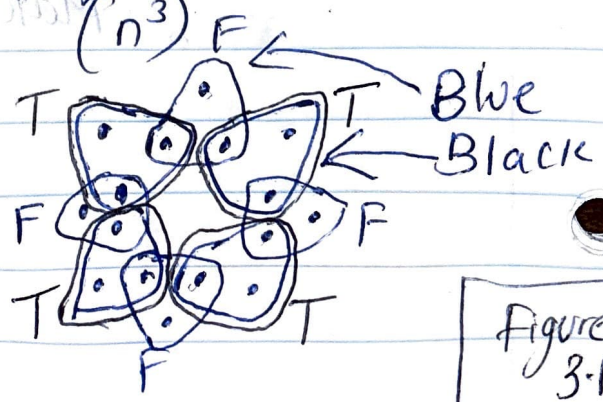


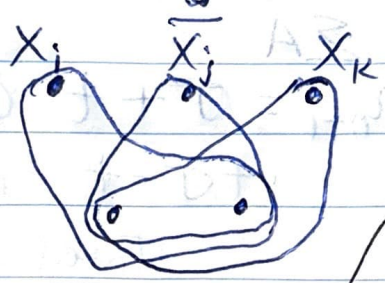
Figure 3.1

let's say Black - True
Blue - False.

$2 \cap X_i$

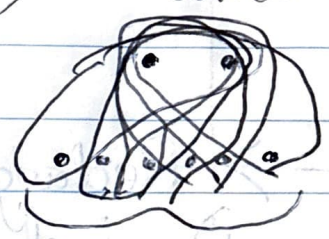
↳ # ocurrences of X_i in the formula (clauses)

→ Clause $X_i \vee \overline{X_j} \vee X_k$



from figure 3.1

Garbage Collection



all T & F dots

$X \sum_{X} n_X$ - No. of clauses.

→ <u>Subset sum</u> :

given n integers $A = \{a_1, a_2 \ldots a_n\}$ and target sum t is there a Subset $S \subseteq A$ Such that

$$\sum S = \sum_{a_i \in S} a_i = t$$

- Weakly NP-hard

- View numbers in base $b = 1 + \max n_{X_i}$

Reduction: - triple $(x_i, x_j, x_k) \rightarrow$ 0000|000|00|000

(i, j, k)

$t = 11111111$

$= \sum_i b^i$ $\qquad = b^i + b^j + b^k$

* Value of numbers is exponential that's why its weakly np-hard

→ Partition Problem : given $A = \{a_1, a_2 \dots a_n\}$, is there a subset $S \subseteq A$ such that

$$\sum S = \sum (A - S) = \frac{\sum A}{2}$$

1) Reduction from Subset Sum.

- let $\sigma = \sum A$
- add $a_{n+1} = \sigma + t$ & add $a_{n+2} = 2\sigma - t$

$+\sigma - t$ $\qquad +t$

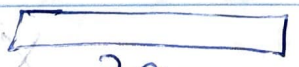
Total $= \sigma$

→ Rectangle packing : R_i →

1) reduction by partition

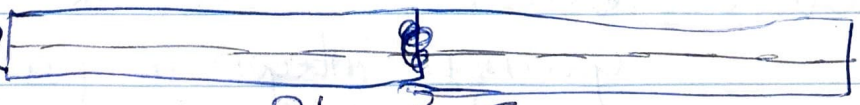
Weakly NP-hard

$a_i \rightarrow$



$3a_i$

$T = 2$

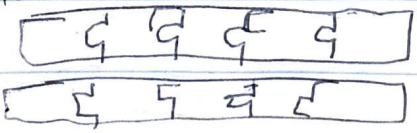


$3t = \frac{3}{2} \sum A$

→ Jigsaw Puzzles

NP-complete

partition does not work



~~Strong~~ Strongly NP-hard

~~4 partion~~

4 partition – Given n integers A, split A into n/4 parts (quadruples) of same ~~size~~ sum

$$\frac{\sum A}{n/4} = t$$

Values $\leq n^{O(1)}$